Anti-Forensic Steganography Using Multi-bit Minimum Error Replacement with Flexible Bit Location

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***Abstract*—Multi-bit minimum error replacement (MER) is a method that can embed multi-bit logo/secret data into *k* least-significant bits (LSBs) of cover data only introduces minimum embedding error (MEE). However, k-LSBs MER suffers from weak anti-forensics. Moreover, it is unfortunate because no literature discusses the formulations and the properties of multi-bit MEE up to this day. Therefore, this work proposes an anti-forensic steganography system using multi-bit MERs with flexible bit location (multi-bit MER_FBL) to overcome the problem of forensics. This work also illustrates the formulations and properties of MEEs in the multi-bit MER_FBL. The embedding error can be easily and determinedly evaluated using the proposed formulations and properties. The proposed multi-bit MER_FBL embeds and hides multi-bit (k-bit, $k \geq 1$) logo/secret data into any adjoining (starting with the ith location) k-bit $b_i b_{i-1} \cdots b_{i-k+1}$ of cover data only introduces MEE in the range of 0 to $\pm(2^k - 1)2^{i-k}$; moreover, its flexible bit location enhances the embedded security as security increases as embedding location increases. As the proposed multi-bit MER_FBL can embed multi-bit logo/secret data into any adjoining bits of cover data and has high embedding capacity and high embedding quality, this method was applied for data steganography. Finally, anti-forensics of the proposed steganography system is demonstrated using the statistical attack of Chi-square analysis.**

Keywords- anti-forensic; steganography; minimum error replacement

I. INTRODUCTION

In information and data hiding, the MER approach is typically used to find data as close to the original as possible. Generally, if one wants to embed k-bit secret data into N-bit ($N \geq k$) cover data, replacing the k-LSBs of cover data will introduce fewer errors than replacing any other k bits. In this case, the embedding error introduced is from 0 to $\pm(2^k - 1)$. Considering N-bit cover data, 2^{N-k} values whose k-LSBs are identical exist in the embedded k bits. To achieve the highest quality, should use the most similar data among these 2^{N-k} values to replace the original. Many LSB-based methods [1-5] have been developed to use in steganography application. Among these studies, A modified EMD method has been proposed in [5] to enhance the embedding capacity because the original EMD has seldom considered embedding capacity. Especially, [5] presented two novel approaches, the HoEMD and the AdEMD, to achieve high efficiency, high quality and large embedding ratios, use the module operation and take into account of the sensitive nature of a human visual system. The HoEMD approach exploits the pixel directions. A pixel with a larger change implies more pixel directions and, ultimately, a larger embedding capacity. The pixel differencing in the proposed AdEMD method is used to evaluate whether the pixel located in the edge area can tolerate a larger change than that of the pixel location in a smooth area. However, disadvantages of all of EMD methods are that have to decide the value of *n* before the hiding process and only concern the k-LSBs embedding.

As all of methods in [1-5] have drawbacks and are not easily implement due to their computational complexity, this work proposes a novel new multi-bit MER_FBL to embed and hide multi-bit (k-bit, $k \geq 1$) logo/secret data into any adjoining (starting with the ith location) k-bit $b_i b_{i-1} \cdots b_{i-k+1}$ of cover data. The proposed multi-bit MER_FBL only introduces minimum embedding error in the range of 0 to $\pm(2^k - 1)2^{i-k}$ by embedding *k* bits and starting at location *i*. It also demonstrates that the flexible bit location enhances system security as security increases as embedding location increases. Moreover, this work develops the formulations and the properties of MEEs to determinedly calculate MEEs based on the multi-bit logo/secret data and the embedding location. The remainder of this paper is organized as follows. Section II presents the formulations and properties of multi-bit bipolar MEEs and the MER_FBL scheme. Section III reports simulation results, demonstrating that the proposed multi-bit MER_FBL scheme is suitable for steganography application, performance comparisons also shown. Finally, section IV gives the discussions and conclusions.

II. THE FORMULATIONS AND PROPERTIES OF MULTI-BIT BIPOLAR MEES AND THE MER_FBL SCHEME

In this section, from the findings of reviewing of MER approaches, we firstly develop the formulations and the

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properties of multi-bit bipolar MEEs to look their features and get more understanding. Next, we will develop the MER_FBL scheme based on the formulations and the properties of multi-bit bipolar MEEs.

A. The Formulations and Properties of Multi-bit Bipolar MEEs

Consider k-bit ($k \geq 1$) secret data should be embedded into $(b_i b_{i-1} \cdots b_{i-k+1})_2$ of N-bit cover data with data value from 0 to $G-1=2^N-1$ where i denotes the most-significant bit (MSB) location of $(b_i b_{i-1} \cdots b_{i-k+1})_2$. We firstly give definitions of some notations in the followings: $P_i = 2^{i+1}$, $N_A = \left\lceil \sum_{m=0}^{k-1} \left(2^{i-m-1}\right) \right\rceil$, $N_B = 2^{i-k+1}$, $N_C = \lceil 2^{i-k} \rceil (2^k+1)$, $BO_ = -\left(\sum_{m=0}^{k-1} \left(2^{i-m}\right) + 1 \right)$, $E_[n] = 2^{i-k+1} - 1 - n$, and $E_+[n] = 2^{i+1} - n$, where notation $\lceil f \rceil$ means the ceiling of f. Next, the k-bit secret data can be divided into two separated groups $\{0x_{k-2}x_{k-3}\cdots x_1x_0\} \leftrightarrow (0x)_2$ and $\{1x_{k-2}x_{k-3}\cdots x_1x_0\} \leftrightarrow (1x)_2$ based on the value of x_{k-1}. Where $x \in \phi$ for $k=1$; otherwise, for $k \geq 2$, x is the binary value of $x_{k-2}x_{k-3}\cdots x_1x_0$. The 1's complementary of x is denoted as $\bar{x} = \overline{(x_{i-1}x_{i-2}\cdots x_{i-k+1})}_2$. Then, the MEE of multi-bit embedding can be expressed as the followings

$$\tilde{e}_{i,k}^{00}[n] = \begin{cases} 0, 0 \leq n \leq N_B - 1 \\ E_[n], N_B \leq n \leq N_C - 1 \\ E_+[n], N_C \leq n \leq P_i - 1 \end{cases}, \; 0 \leq n \leq P_i - 1. \tag{1}$$

$$\tilde{e}_{i,k}^{0x}[n] = \tilde{e}_{i,k}^{00}[(n - xN_B)\%P_i], \; 0 \leq n \leq P_i - 1. \tag{2}$$

$$EP_{i,k}^{0x}[l] = \tilde{e}_{i,k}^{0x}[l\%P_i], \; 0 \leq l \leq G-1. \tag{3}$$

$$BO_{i,k}^{0x}[l] = \begin{cases} 0, 0 \leq l \leq G - N_A + xN_B - 1 \\ BO_, G - N_A + xN_B \leq l \leq G-1 \end{cases}. \tag{4}$$

$$e_{i,k}^{0x}[l] = EP_{i,k}^{0x}[l] + BO_{i,k}^{0x}[l], \; 0 \leq l \leq G-1. \tag{5}$$

$$e_{i,k}^{1x}[l] = -e_{i,k}^{0\bar{x}}[G-1-l], \; 0 \leq l \leq G-1. \tag{6}$$

A special sequence $\tilde{e}_{i,k}^{00}[n]$ is defined in Eq. (1) and it's right rotation version $\tilde{e}_{i,k}^{00}[(n - xN_B)\%P_i]$ with xN_B steps in the range of P_i, that is $\tilde{e}_{i,k}^{0x}[n]$, is defined in Eq. (2). Notably both $E_[n]$ and $E_+[n]$ of $\tilde{e}_{i,k}^{00}[n]$ are mutually anti-symmetric. Either $\tilde{e}_{i,k}^{00}[n]$ or $\tilde{e}_{i,k}^{0x}[n]$ is the fundamental sequence of the periodic sequence $EP_{i,k}^{00}[l]$ or $EP_{i,k}^{0x}[l]$ with period P_i which have been defined in Eq. (3). We called $EP_{i,k}^{0x}[l]$, $0 \leq l \leq G-1$ as *periodic error sequence* due to it is periodic and is one of major components of MEE. The BO sequence of $(0x)_2$ group is denoted as $BO_{i,k}^{0x}[l]$ and is defined by Eq. (4). It shows that the non-zero value of $BO_{i,k}^{0x}[l]$ and its starting point depend on the values of both i and k. Moreover, the value and the length of non-zero value of $BO_{i,k}^{0x}[l]$ increase as the values of i and/or k increases. That is, higher embedding location and/or more embedding bits will introduce bigger magnitude and longer length of BO than those of lower embedding location and/or less embedding bits. Eq. (4) also shows that there are 2^{k-1} sequences of $BO_{i,k}^{0x}[l]$ with the same length G in the group $(0x)_2$ but their length of non-zero value are different. It is clear that the length of non-zero value of $BO_{i,k}^{0x}[l]$ is decreasing with value of N_B as the value of x increasing. It also clearly shows non-zero values of $BO_{i,k}^{0x}[l]$ are negative and appears in around of G-1. Eq. (5) means that MEE $e_{i,k}^{0x}[l]$ consists of the periodic error sequence $EP_{i,k}^{0x}[l]$ and the boundary overshoot $BO_{i,k}^{0x}[l]$. Eq. (6) shows that MEE $e_{i,k}^{1x}[l]$ is negative and position-reversal of $e_{i,k}^{0\bar{x}}[l]$. Because $e_{i,k}^{1x}[l]$ is negative and position-reversal of $e_{i,k}^{0\bar{x}}[l]$ therefore it can be evaluate from $e_{i,k}^{0\bar{x}}[l]$ even if $BO_{i,k}^{1x}[l]$ and $\tilde{e}_{i,k}^{1x}[n]$ are unknown.

For detailed illustration, we give an example of MEEs for embedding 2-bit (k = 2) secret data into b_5b_4 (i = 5) of 8-bit cover data where P_i = 64, G = 256, $N_A = 24$, $N_B = 16$, $N_C = 40$, $BO_ = -49$, $E_[n] = 15 - n$ and $E_+[n] = 64 - n$, and 8-bit cover data has the value $0 \leq l \leq G-1 = 2^8 - 1 = 255$. The 2-bit secret data can be divided into $(0x)_2 = \{00_2, 01_2\}$ and $(1x)_2 = \{10_2, 11_2\}$. Using Eqs. (1), (2), (3), (4) and (5), we have $\tilde{e}_{5,2}^{0x}[n]$, $EP_{5,2}^{0x}[l]$, $BO_{5,2}^{0x}[l]$, and $e_{5,2}^{0x}[l]$, which are shown in Figure 1.

Figure 1 shows all figures of the example of MEE for '0x'-embedding which embeds '0x' into b_5b_4 of 8-bit cover data. Figure 1a shows the figures of fundamental sequences $\tilde{e}_{5,2}^{0x}[n], 0 \leq x \leq 1$ which are generated from Eqs. (1) and (2). Notably, in Figure 1a, it clearly finds that $\tilde{e}_{5,2}^{01}[n]$ is the right rotation of $\tilde{e}_{5,2}^{00}[n]$ with 16 steps as the expression of Eq. (2). Figure 1b displays the figures of periodic error sequences $EP_{5,2}^{0x}[l], 0 \leq x \leq 1$, it is clear that $EP_{5,2}^{0x}[l], 0 \leq x \leq 1$ are periodic version of $\tilde{e}_{5,2}^{0x}[n], 0 \leq x \leq 1$ as the expression of Eq. (3), in the range from 0 to 255. We also find that $EP_{5,2}^{01}[l]$ is the right rotation of $EP_{5,2}^{00}[l]$ with 16 steps. Figures of boundary overshoots $BO_{5,2}^{0x}[l], 0 \leq x \leq 1$ are shown in Figure 1c. It clearly shows that there are 24 negative values (-49) appear nearby 255 (the maximum value G-1) of cover data for $BO_{5,2}^{00}[l]$, which are given by Eq. (4). Whereas, there are only 8 negative values (-49) appear nearby 255 of cover data for $BO_{5,2}^{01}[l]$. This fact means that $BO_{5,2}^{00}[l]$ is worse than $BO_{5,2}^{01}[l]$. The MEEs for embedding 2-bit secret data '0x' into b_5b_4 of 8-bit cover data are summarized in Figure 1d. It is clear that

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the MEEs are summation of the periodic error sequence and the boundary overshoot as Eq. (5). Notably, Figure 1d shows that the MEE of '00'-embedding is worse than that of '01'-embedding because '00'-embedding suffers from worse boundary overshoot than '01'-embedding.

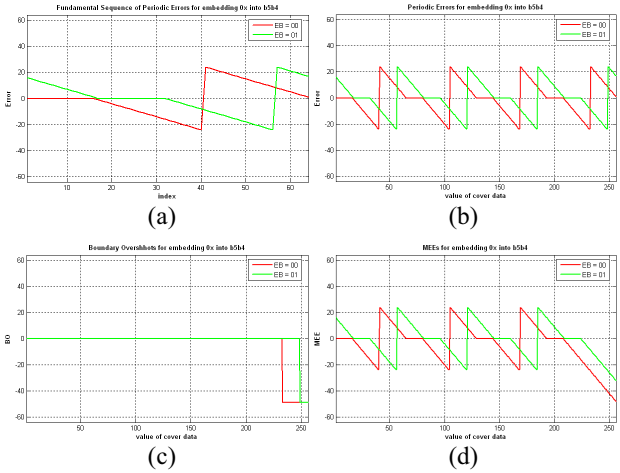

Figure 1. Illustrations of MEE for '0x'-embedding which embeds '0x' into b_5b_4 of 8-bit cover data. (a) Fundamental sequences $\tilde{e}_{5,2}^{0x}[n], 0 \le x \le 1$, (b) Periodic error sequences $EP_{5,2}^{0x}[l], 0 \le x \le 1$, (c) Boundary overshoots $BO_{5,2}^{0x}[l], 0 \le x \le 1$, (d) Minimum embedding errors $e_{5,2}^{0x}[l], 0 \le x \le 1$.

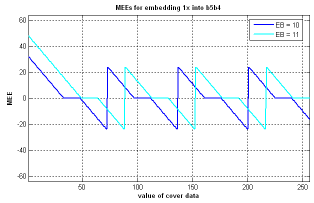

Figure 2. Illustrations of MEEs $e_{5,2}^{1x}[l], 0 \le x \le 1$ for '1x'-embedding which embeds '1x' into b_5b_4 of 8-bit cover data.

Using Eq. (6), we have $e_{5,2}^{1x}[l]$, which are depicted in Figure 2. It is clear that the MEEs are summation of the periodic error sequence and the boundary overshoot as Eq. (6). Notably, Figure 2 shows that the MEE of '11'-embedding is worse than that of '10'-embedding because '11'-embedding suffers from worse boundary overshoot than '10'-embedding. Figures 1 and 2 clearly show that the MEE sequences for '$0x$'-embedding and those for '$1\bar{x}$'-embedding are mutually negative and position-reversal in the defined domain from 0 to 255.

B. *The MER_FBL scheme*

We have developed the MER_FBL scheme based on the formulations and the properties of multi-bit bipolar MEEs as in Figure 3. The block of fundamental sequence generator firstly generates the fundamental sequences based on Eqs. (1), and (2) under the controls of the starting location *i*, bits of cover data *N*, and *k*-bit secret data (includes bits of secret data k). Next, the periodic error generator produces the periodic sequences using the fundamental sequences under the controls of the starting location *i*, bits of cover data *N*, and *k*-bit secret data (includes bits of secret data k) as the indication of Eq. (3). At the same time, boundary overshoot generator yields sequences of boundary overshoot based on Eq. (4) under the controls of the starting location *i*, bits of cover data *N*, and *k*-bit secret data (includes bits of secret data k). Then, the MEE generator produces the MEE sequences. Finally, the MER_FBL scheme adds the cover data and the MEE data to produce the embedded data.

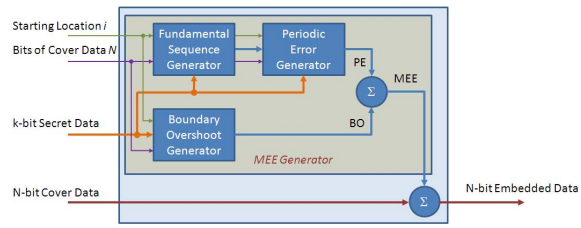

Figure 3. MER_FBL scheme for hiding data.

III. Simulation Results and Performance Comparisons

Many simulations of steganography were conducted to test performances of the proposed MER_FBL scheme.

A. *Simulation Results*

Note that all cover images are true color (24-bit) and with size of 256×256 in our simulations. Besides, all secret images are 1-bit (monochrome), 2-, 3-, 4-, 5-, 6-, 7-, 8-bit index color versions and sized 256×256. Table 1 shows many tested cover images with different PSNRs and embedding capacities (ECs). Clearly the proposed stegangraphy scheme achieved high-quality stego-images even when EC achieved 50%. During experiments, we knew that retrieved secret images were lossless because they are identical to the original images. The proposed MER_FBL scheme generates good-quality stego-images for steganography.

TABLE 1 PSNR for cover images with different ECs

Cover Image	EC (%)				
	12.5	25.0	37.5	50.0	62.5
Baboon	51.12	46.40	40.72	34.76	28.74
Barbara	51.15	46.40	40.77	34.92	28.92
Boat	51.16	46.36	40.73	34.95	28.75
Goldhill	51.16	46.36	40.72	34.78	28.87
Jet	51.14	46.38	40.65	34.74	28.74
Lena	51.13	46.37	40.75	34.89	28.72
Sailbot	51.13	46.41	40.76	34.76	28.69
Tiffany	51.48	46.35	40.73	34.79	28.13
Average	51.17	46.38	40.69	34.72	28.62

B. *Performance Comparisons*

Table 2 shows performance comparisons between the proposed multi-bit MER_FBL scheme and other methods. According to comparison results, the proposed multi-bit MER_FBL scheme performs better than those of [1-5].

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TABLE 2 Comparisons with other works

Images	Lena/Sailboat				
EC	33.3%	37.5%	45.6%	50%	54.2%
[1]	-/-	-/-	-/-	32.57/-	-/-
[2][a]	-/-	-/-	-/-	33.10/-	-/-
[3]	-/-	-/-	-/-	34.90/-	-/-
[4]	-/-	37.91/37.93	-/-	31.84/31.72	-/-
[5]-1[b]	-/-	38.16/38.08	-/-	34.38/34.29	
[5]-2[c]	-/-	38.98/38.85	-/-	33.10/33.54	-/-
Our	41.92/41.92	40.75/40.76	36.00/36.07	34.89/34.76	31.79/31.78

a. The PSNR value is averaged value of the four PSNR values that were shown in Fig. 13 of [2].
b. [5]-1 means HoEMD method.
c. [5]-2 means AdEMD method.

C. *Security Analysis*

Security is a vital concept for developing a data hiding scheme, thus a statistical attack of Chi-square analysis [6] is used to demonstrate that the proposed multi-bit A3S embedding algorithm can satisfy the security requirements. Three curves as displayed in Figure 4, including the red one (the bottom line in the simulation result), the green one (distributed in middle of the simulation result), and the blue one (the top line in the simulation result), are depicted in the results of the Chi-square test. In this analysis, the red curve indicates the outcome of the Chi-square test. In general, the red curve is reclined on the line of zero when a natural image is used as the test image; in other words the object has a high probability of embedding random messages if the red curve is close to one. Namely, secret data have been embedded in this object. The second output is the blue curve, in which each vertical blue line represents 1K bytes of embedded data. The third output is the green curve that remains around 0.5 if a random message is embedded. Figures in are summarized as the undetected results since the red curves are close to zero. According to these results, the proposed approach is undetected by using Chi-square testing as good as in previous studies of EMD, HoEMD and AdEMD.

Figure 4. Chi-square tests of Lena with EC 33.3% and 50%.

IV. Discussions And Conclusions

This work develops a novel multi-bit MER_FBL scheme based on the formulations and the properties of MEE for hiding data. The proposed multi-bit MER_FBL scheme can embed k-bit ($k \geq 1$) secret data into cover data only introduce minimum embedding error to be restricted to $\pm\left(2^{k-1}-1\right)2^{i-k}$ according to the embedding location *i*. Many simulations show that the proposed multi-bit MER_FBL scheme performs good embedding quality for steganography application. Reasons that we develop the proposed multi-bit MER_FBL are (1) to overcome the drawbacks that exist in the previous works and (2) to develop the formulations and the properties of MEE to determinedly and easily calculate MEEs based on the multi-bit logo/secret data and the embedding location.

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